RNA-SEQ DIFFERENTIAL EXPRESSION

Master Runbook

Airway Smooth Muscle / Dexamethasone Study (GSE52778)

Project	Differential_Gene_Expression_Study (R / DESeq2)
Environment	R + Bioconductor, run via RStudio project
Curriculum	MIBO8110 Applied Omics
Document type	Operational Standard Operating Procedure (SOP) Runbook
Companion document	README.md (repository overview, architecture, citation)
Companion repositories	mibo8110_First_R_Exercise, Automated_validation_function

Generated for internal course and reproducibility use.

0. Document Control

Field	Value
Scope	Executing the DESeq2 differential-expression workflow, scripts 00 through the extended visualization/optional-analysis stage
Audience	Students and instructors running the analysis in R/RStudio
Companion doc	README.md (architecture, citation, license, data background)
Verification status	Reviewed line-by-line; one data-integrity issue found and fixed during this review (stale aligned CSVs) — see §8

1. Purpose & Scope

This Runbook is the step-by-step operational companion to the repository's README.md. It gives exact commands, what to expect at each checkpoint, and what to do when a step fails. It assumes R and RStudio are installed, with Bioconductor package-installation access.

The workflow starts from an already-quantified gene × sample count matrix (not raw FASTQ reads) and carries it through validation, a DESeq2 model fit, a treatment-vs-control contrast, and a full set of standard RNA-seq diagnostic and results plots.

Read this before you start

Section 8 documents a real data-integrity issue found in this repository's shipped output files (a stale, incomplete counts_aligned.csv) and how it was fixed. If you already have your own copy of this repository from before this cleanup, regenerate data/counts_aligned.csv and data/metadata_aligned.csv by re-running Aligning_meta_counts.R rather than trusting old copies.

2. Study & Data Background

Field	Value
Study	Himes BE, et al. (2014). RNA-Seq Transcriptome Profiling Identifies CRISPLD2 as a Glucocorticoid Responsive Gene that Modulates Cytokine Function in Airway Smooth Muscle Cells. PLOS ONE.
DOI	10.1371/journal.pone.0099625
GEO series	GSE52778
Design	4 donor cell lines × 2 conditions (dexamethasone-treated vs. untreated) = 8 samples, paired by donor
Input files	`data/counts.csv` (gene × sample counts), `data/metadata2.csv` (sample ID, cell line, treatment)
Statistical design	`~ cellline + dexamethasone` — adjusts for donor baseline variation while estimating the treatment effect

3. Repository Architecture

This is one of three related, separately published repositories:

- **Differential_Gene_Expression_Study** (this repo) — the full analysis: validation, DESeq2 model, core plots, and an extended visualization/optional-analysis stage (shrinkage, GO enrichment/GSEA, KEGG pathway overlay).
- **mibo8110_First_R_Exercise** — a compact starter version of the same exercise for a student's first R + GitHub project.
- **Automated_validation_function** — the counts/metadata validation function used in step 3 below, published standalone with its own example data.

Package management (00), directory setup (01), and file organization (02) are shared, near-identical scripts across the first two repositories by design — they are simple, reusable project scaffolding, not something that needed deduplicating across separately published repos.

4. Prerequisites

4.1 Software

- R, a recent 4.x release, with Bioconductor access.
- RStudio (recommended) — opening the `.Rproj` file sets the working directory to the project root automatically, which every script assumes.

4.2 R packages

Package	Source	Category	Purpose
DESeq2	Bioconductor	Core	Differential-expression analysis
ggplot2	CRAN	Core	General plotting
dplyr	CRAN	Core	Data manipulation
tidyr	CRAN	Core	Reshaping data for plots
pheatmap	CRAN	Core	Heatmaps
RColorBrewer	CRAN	Core	Heatmap color palettes
apeglm	Bioconductor	Optional	Log2 fold-change shrinkage
clusterProfiler	Bioconductor	Optional	GO enrichment and GSEA
enrichplot	Bioconductor	Optional	Enrichment plots
org.Hs.eg.db	Bioconductor	Optional	Human gene annotations
pathview	Bioconductor	Optional	KEGG pathway diagrams

Run `scripts/00_manage_project_packages.R` and call `manage_packages("automatic")` to install everything above in one step; call `manage_packages("list")` (the default) just to check status without installing anything.

5. Directory Structure

```
Differential_Gene_Expression_Study/
├── Differential_Gene_Expression_Study.Rproj
├── 02_organize_project_files.R      # deliberately also at the root — see README
├── scripts/   (00_.. through Additional_high-value_..R)
├── data/      (counts.csv, metadata2.csv, *_aligned.csv, cnt_adj.csv, met_adj.csv)
├── plots/     (hsa04110.png, hsa04110.pathview.png, hsa04110.xml)
└── docs/      (class exercise handout, troubleshooting notebook, install notes)
```

6. Standard Operating Procedure

Open `Differential_Gene_Expression_Study.Rproj` in RStudio first. Every command below assumes the working directory is the project root.

Step 0 — Package status and installation

```
source("scripts/00_manage_project_packages.R")
manage_packages("automatic")
```

- What it does: prints an install-status table for every package this project uses, then installs (CRAN and Bioconductor) whatever is missing.
- Verify: re-run `manage_packages("list")` and confirm the printed count reads N of N packages installed.
- Alternative: `scripts/00_install_and_load_packages.R` shows the same two installation approaches (manual, automated) spelled out individually — useful as a teaching reference, not required if step 0 above already succeeded.

Step 1-2 — Project directories (already applied in this repository)

```
source("scripts/01_create_project_directories.R")
source("02_organize_project_files.R")   # run from the project root, not from inside scripts/
```

- What it does: creates `scripts/`, `data/`, `plots/` if missing, then moves any loose `.R`/`.csv`/plot files at the project root into them. Both are idempotent — safe to re-run, and existing destination files are never overwritten.
- This repository is already organized this way; you only need this step if you're starting from a fresh, flat copy of the raw files.

Step 3 — Validate and align counts + metadata

```
source("scripts/Aligning_meta_counts.R")
```

- What it does: reads `data/counts.csv` and `data/metadata2.csv`, correctly handling the trailing-comma header (see §7), validates sample IDs (no duplicates, no mismatches), reorders metadata to match the count columns, and writes `data/counts_aligned.csv` + `data/metadata_aligned.csv`.
- Verify: console prints `SUCCESS: aligned 8 samples and 64102 genes.` — if the sample count is not 8, or the script stops with a specific error, see §9.

- Under the hood this uses the same checks as `validate_rnaseq_inputs()` in the companion Automated_validation_function repository.

Step 4 — Build the DESeq2 model and run the contrast

```
source("scripts/LOAD_CLEANED_ALIGNED_DATA.R")
```

- What it does: loads the aligned pair, builds a `DESeqDataSet` with design `~ cellline + dexamethasone`, filters genes with fewer than 10 total reads across all samples, fits the model, and extracts the `dexamethasone: treated vs. untreated` contrast at alpha = 0.05.
- Verify: console prints `SUCCESS: DESeq2 analysis completed for <N> genes.`, followed by `summary(res)` and the top genes by adjusted p-value.
- If it stops with a sample-order or column-name error, re-run Step 3 first — this script explicitly checks that `metadata_aligned.csv` still has `cellline` and `dexamethasone` columns in the same order as the count columns.

Step 5 — Core results plots

```
source("scripts/VISUALIZE_DESeq2_RESULTS.R")
```

- What it does: PCA plot (colored by treatment, shaped by cell line), a volcano plot (padj < 0.05, |log2FC| > 1), and a boxplot of the single most significant gene's normalized counts.
- Requires: Step 4's `dds` and `res` objects to already exist in your R session.

Step 6 — Extended QC/results plots and optional analyses

```
source("scripts/Additional_high-value_visualization_plots_commonly_used_in_RNA-seq_analyses.R")
```

- What it does unconditionally: a sample-to-sample distance heatmap, a top-30 significant-gene heatmap (row Z-scores), an MA plot, a raw p-value histogram, a DESeq2 dispersion diagnostic, and top-10-gene expression profile lines faceted by cell line.
- Optional (edit `run_lfc_shrinkage` / `run_enrichment` / `run_pathview` at the top of the script, all `FALSE` by default): apeglm log2FC shrinkage; GO enrichment (`enrichGO`) and GSEA (`gseGO`) against org.Hs.eg.db; a KEGG pathway overlay via pathview (defaults to hsa04110, Cell cycle — the `plots/hsa04110.*` files in this repo are the output of that run).

> **pathview writes files to your working directory**
>
> The pathview call uses `kegg.dir = "plots"`, so run this from the project root (as instructed above) or its downloaded KGML/PNG files will land in the wrong place.

7. Data Quirks This Pipeline Defends Against

7.1 Trailing comma in counts.csv's header

The header row ends in a comma, which R reads as a spurious extra column named `X` once default column-naming is applied — and a naive `read.csv()` also risks misaligning the very first sample column (SRR1039508) against the header. `Aligning_meta_counts.R` reads the header line separately with `scan()` and reconstructs column names explicitly rather than trusting `read.csv()`'s own header parsing.

7.2 Sample IDs are an ordinary column in metadata2.csv, not row names

A naive `read.csv()` leaves `rownames(met)` as `"1"` through `"8"` (R's default integer row labels), which silently fails `all(colnames(cnt) %in% rownames(met))` in a way that is not obvious from `head()` alone — the printed table looks fine either way. `Aligning_meta_counts.R` promotes that first column to real row names and validates the result explicitly.

Both quirks are documented step by step, with the original diagnostic commands and their actual output, in `docs/RNAseq_Troubleshooting_Notebook.Rmd` and in `scripts/DESeq.R` (the original live debugging session).

8. Known Issues (Fixed During This Cleanup)

8.1 Stale, incorrect aligned-data outputs

`data/counts_aligned.csv` and `data/metadata_aligned.csv`, as originally shipped in this repository, were the output of an earlier, since-corrected version of the alignment logic:

- `counts_aligned.csv` was missing the SRR1039508 column entirely — 7 samples instead of 8.
- `metadata_aligned.csv`'s columns were mislabeled: `"dexamethasone","X"` instead of `"cellline","dexamethasone"`.

Both files have been replaced with correct, verified output — cross-checked against an independently generated identical copy from the companion `mibo8110_First_R_Exercise` repository, and consistent with what the current `Aligning_meta_counts.R` produces from the raw inputs. If you re-run `Aligning_meta_counts.R` yourself, it will regenerate both files identically to what is now committed here.

Practical impact if uncorrected

`LOAD_CLEANED_ALIGNED_DATA.R` trusts `metadata_aligned.csv`'s column names directly (`required_columns <- c("cellline", "dexamethasone")`) — against the old file it would have stopped with a clear error rather than silently modeling the wrong variable, but with one fewer sample and a hard failure either way. Re-running Step 3 (`Aligning_meta_counts.R`) before Step 4 always regenerates a correct pair regardless.

8.2 Syntax error in the historical install notes

`docs/Bioconductor_Pacakage_Installation.txt` had a missing closing parenthesis on `BiocManager::install(c("GenomicFeatures")` that would have stopped a copy-pasted install partway through. Fixed.

No script logic beyond these two fixes was changed during this cleanup.

9. Troubleshooting Reference

Distilled from `docs/RNAseq_Troubleshooting_Notebook.Rmd`.

9.1 If sample validation fails

```
# Inspect structure, not just appearance
dim(cnt); dim(met)
str(cnt); str(met)
colnames(cnt); rownames(met)

# Which side has extra/missing samples?
setdiff(colnames(cnt), rownames(met))
setdiff(rownames(met), colnames(cnt))
```

9.2 General data-import best practices

- Keep canonical data in CSV/TSV, not XLSX; preserve an untouched source copy.
- Use one row per sample, one column per metadata variable, with a unique non-empty sample ID as the key.
- Keep sample IDs identical across counts, metadata, and any filenames.
- Never rely on a printed `head()` alone — inspect `dim()`, `str()`, `rownames()`, `colnames()` explicitly.
- Fail early with `stopifnot()` or a validation function rather than proceeding on an unverified assumption.
- Record package versions with `sessionInfo()` alongside any results you report.

9.3 Final pre-analysis checklist

```
stopifnot(!anyDuplicated(colnames(cnt)))
stopifnot(!anyDuplicated(rownames(met)))
stopifnot(setequal(colnames(cnt), rownames(met)))
met <- met[colnames(cnt), , drop = FALSE]
stopifnot(identical(colnames(cnt), rownames(met)))
stopifnot(all(vapply(cnt, is.numeric, logical(1))))
stopifnot(!anyNA(cnt))
stopifnot(all(as.matrix(cnt) >= 0))
```

10. Downstream / Optional Analyses

All three are off by default in `scripts/Additional_high-value_visualization_plots_commonly_used_in_RNA-seq_analyses.R` — set the corresponding flag to `TRUE` after installing the packages listed in §4.2.

10.1 Log2 fold-change shrinkage (apeglm)

Reduces noisy fold-change estimates, especially for low-count genes. Produces a side-by-side unshrunken vs. shrunken MA plot.

10.2 GO enrichment and GSEA (clusterProfiler)

`enrichGO()` tests significant genes (padj < 0.05) for GO Biological Process enrichment; `gseGO()` ranks all tested genes by test statistic and runs gene set enrichment without a significance cutoff.

10.3 KEGG pathway overlay (pathview)

Maps log2 fold changes onto a named KEGG pathway diagram (defaults to hsa04110, Cell cycle). The `plots/hsa04110.png`, `hsa04110.pathview.png`, and `hsa04110.xml` files in this repository are the output of a completed run.

Appendix A — Full Command Reference

```
# Open Differential_Gene_Expression_Study.Rproj in RStudio first
source("scripts/00_manage_project_packages.R"); manage_packages("automatic")
source("scripts/Aligning_meta_counts.R")
source("scripts/LOAD_CLEANED_ALIGNED_DATA.R")
source("scripts/VISUALIZE_DESeq2_RESULTS.R")
source("scripts/Additional_high-value_visualization_plots_commonly_used_in_RNA-seq_analyses.R")
```

Citation

Himes BE, Jiang X, Wagner P, Hu R, Wang Q, Klanderman B, et al. (2014) RNA-Seq Transcriptome Profiling Identifies CRISPLD2 as a Glucocorticoid Responsive Gene that Modulates Cytokine Function in Airway Smooth Muscle Cells. *PLoS ONE* 9(6): e99625. https://doi.org/10.1371/journal.pone.0099625